Identification of an Unknown Sample B

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INTRODUCTION

The objective of the lab was to correctly identify an unknown molecule, sample B, using data from Mass Spectroscopy, Infrared Spectroscopy and NMR graphs. All three of these methods are used in organic chemistry to analyze and identify unknowns. Each graph gives a unique form of the molecules properties that when used together can help determine the unknown. The Mass Spectroscopy graph is used to determine ions present in the molecule and tells the molecule's molar mass. The Infrared Spectroscopy graph shows what functional groups are present in the molecule. This is achieved by using infrared waves to excite bonds as they absorb infrared radiation. The Nuclear Magnetic Resonance Graph is used to find out how many hydrogen atoms are in the molecule as well as where they are distributed. All of this information is then interpreted and integrated to determine the molecule.

THEORY

Infrared Spectroscopy:

Infrared spectroscopy generates a spectrum by measuring the bonds within a molecule and how they react to infrared radiation. Some bonds vibrate when exposed to infrared spectroscopy. These vibrations are measured. The amplitude increases if the radiation frequency is the same as the bond's radiation. Based on the molecule's shape, different patterns are produced because they absorb the radiation at different places. The two main regions of this spectrum in the functional group region and the fingerprint region. Some functional groups may fall in the fingerprint region of 1200 *cm-1* to 600 *cm-1*, most function groups reside in the functional group region from 3600 *cm-1* to 1200 *cm-1.* Because single, double, and triple bonds absorb different amounts, they are seen at different locations of the graph. The range for single bonds formed between carbon, hydrogen, oxygen, and nitrogen is 3600 *cm-1* to 2850 *cm-1.* Therefore if peaks appear in this area it is most likely a single bond. The range of triple bonds between carbon and nitrogen is 2300 $cm{-}1$ to 2100 $cm{-}1$, so most alkynes will be found in this region. Alkenes will be found in the region containing most double bonds between carbon, oxygen and nitrogen which is the region of 1850 $cm{-}1$ to 1620 $cm{-}1$. The instrument IR spectrometer is used to excite the bonds and absorb their frequencies. The sample is put in the apparatus and exposed to the IR light. The amount of light absorbed is recorded and an infrared plot showing the light frequencies is created. Since each signal has a corresponding structure or group, the molecule's structure can be determined.

Mass Spectroscopy:

Mass spectroscopy analyzes particles based on their molecular weight and mass. The mass spectrometry process involves four key steps. Initially, through electron impact ionization, the organic molecule undergoes a conversion into positive ions within the apparatus by shedding one or more electrons. Subsequently, a high-energy electron beam bombards the molecule in the gas phase within the machine. This results in the formation of an intermediate entity known as a molecular ion (M)+, characterized by being both a radical and a high-energy cation. In the second stage, all cations are uniformly accelerated to possess the same kinetic energy. During the third stage, a magnetic field deflects each cation based on its mass. Lighter and more charged ions experience greater deflection. If a significant number of electrons are lost in the initial stage, the ion becomes strongly positively charged and is deflected accordingly. The detector then provides precise information on the ion size and signal intensity. A thorough examination of the mass spectrum uncovers multiple notable peaks, with the base peak being the foremost. Representing the most prevalent fragment ion at 100% relative abundance, the base peak offers insights into the molecule's dominant features. Variations in the peaks indicate additional characteristics of the molecule, enabling the identification of the constituent elements. Elements with numerous isotopes exhibit multiple isotope peaks in their spectrum, contributing to a comprehensive analysis.

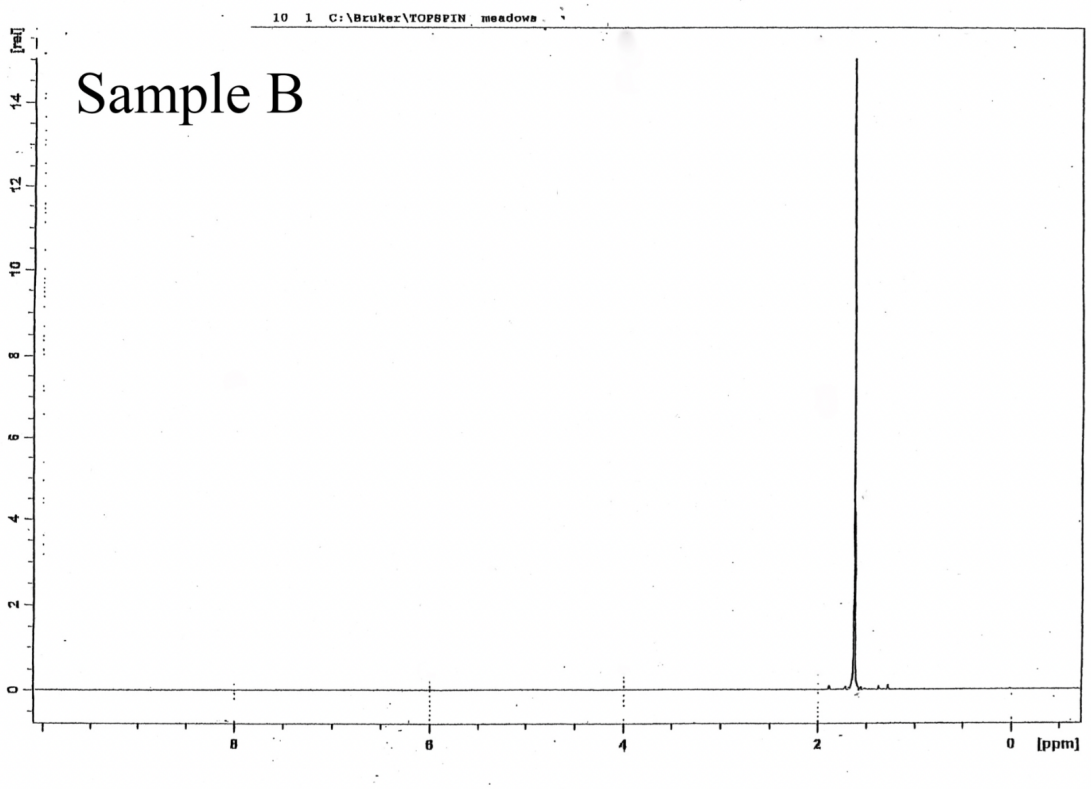
NMR:

Nuclear magnetic resonance (NMR) facilitates the precise determination of atom configuration and connectivity within a molecule. It allows for the identification of specific functional groups and hydrogen locations, crucial for understanding a molecule's carbon and hydrogen framework,

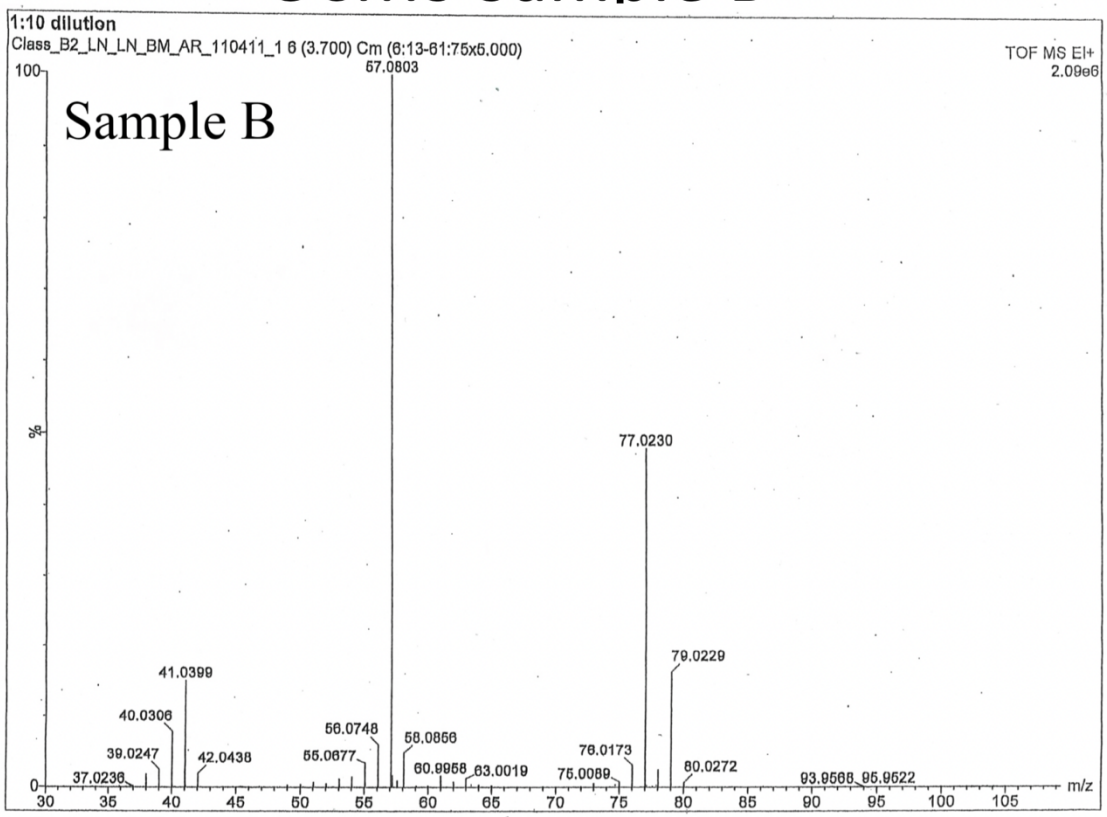
thereby aiding in its identification. NMR involves the study of nuclei and their reactions through electromagnetic radiation. When subjected to a radio frequency within a magnetic field, nuclei shift to a higher energy state. Protons achieve resonance when the energy difference is negligible. NMR utilizes low radio frequency waves, and a spectrum is generated after exposing the sample to the NMR machine. Analyzing an NMR spectrum involves four key considerations. Firstly, the number of signals on the spectrum is examined. Each specific signal provides insights into the electronic environment of a proton and its interactions within the molecule. A signal may represent a single proton or multiple protons sharing the same signal due to a shared electronic environment. Secondly, the signal's location is crucial, defined by the difference between the proton's absorption frequency and the spectrometer's standard absorption, known as a chemical shift. The magnitude of the chemical shift reflects the proton's location within the molecule, with shielded protons closer to zero on the right side and deshielded protons closer to ten on the left side of the spectrum. The third aspect involves examining the region beneath the signal. Integration helps calculate the number of protons in the signal, providing information about the signal's area. Finally, the fourth consideration is the number of peaks at each signal, known as multiplicity. Multiplicity, determined by the formula n+1, is influenced by the number of protons on the nearby carbon atom. Understanding the relationship between carbons and their connections within the molecule is facilitated by multiplicity.

DATA

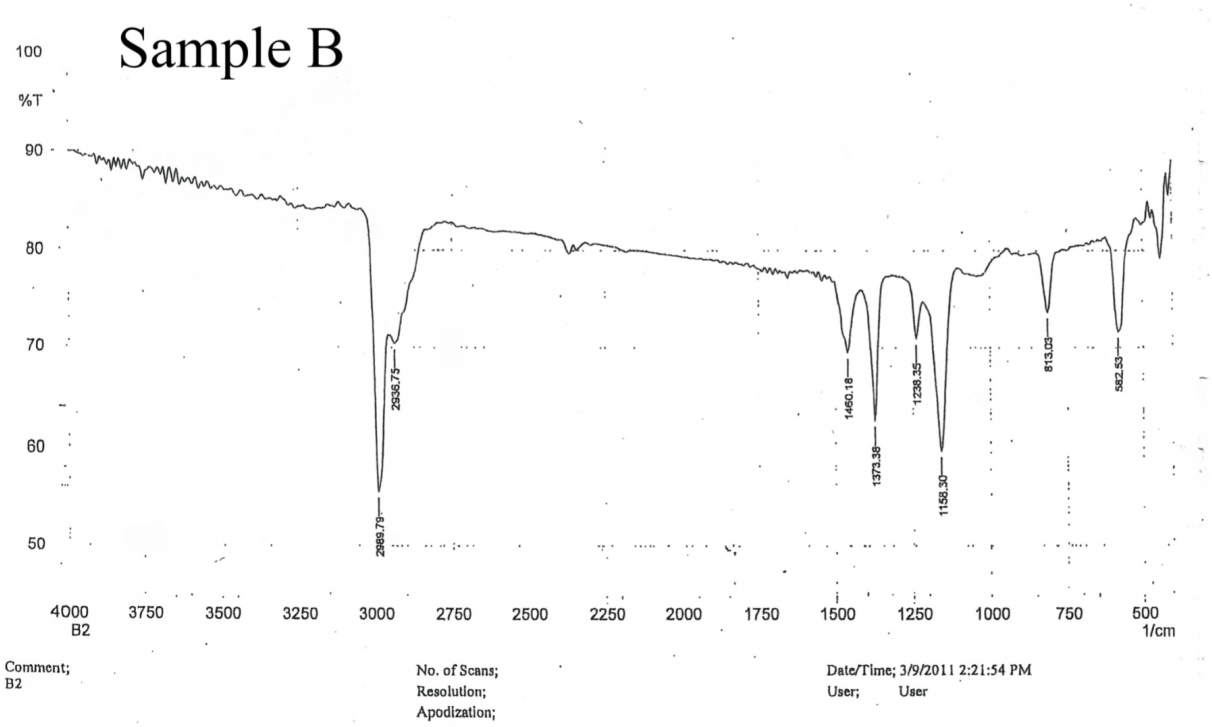
NMR Sample B



GCMS Sample B



FTIR Sample B



ANALYSIS

FTIR:

There is no activity above 3000 *cm-1.* This means there is no alcohol group within the compound. The peaks at 2989.79 and 2936.75 indicate that there are C-H bonds in the compound. Also, there is no peak at 1600-1700 *cm-1*, which means there is no double or triple bonds. There are some signals between 1500-100 *cm-1*, which identify methyl group. The peaks at 813.03 and 582.83 tells that there is a C-Cl bond.

NMR:

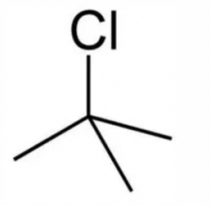
The graph shows a single peak at about 1.6 ppm and the primary protons peaks at 1.11.2 ppm. This means there is a chloride atom, and it causes a chemical downshift. The presence of single peak tells that there is only one type of proton present in the compound. Since the signal is tall, there must be multiple protons giving rise to this signal.

GCMS:

The base peak is at 57 m/z, this means that the most stable and possible cation chemical formula is C4H9. (M+1) + peak is at 77 and (M+2) + peak is at 79, in which (M+2)+ is about 1/3 the height of (M+1)+, which indicate the chlorine atom. The molecular ion peak is found at 93, which indicates the chemical formula of tert-Butylchloride.

CONCLUSION

Using the three methods and analysis, it can be concluded that the unknown chemical compound B is tert-Butylchloride (1-chloro-3-methylpropane). The chemical structure of it is shown below.



REFERENCES

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